

Is It Possible to Make Our Own AI/NLP Model?

ExamiQ+ thesis guidance — feasibility of custom models vs hosted LLM inference

Yes in a limited sense — no as in “train our own GPT from scratch.”

What’s *not* realistic for your thesis

Idea	Why not
Train GPT/BERT from scratch	Needs huge data, GPUs, months, ML research skillset
Fine-tune a big model on all student data	Privacy/ethics, small N, weak results, hard to defend
Match Ollama/Gemini quality alone	Commercial models win; you’d underperform and spend the whole thesis on ML

That’s why ExamiQ correctly uses a hosted LLM for generation and tutoring.

What *is* possible (panel-friendly as “our model”)

You can claim a **custom, domain-scoped model** if you mean one of these:

- 1. Rule-based / hybrid “model” (you already have pieces).** Explanation steps, calibration logic, adaptive selection, stubs when AI is off — this is *your* educational intelligence, not ChatGPT.
- 2. Small fine-tune or classifier (realistic add-on).** e.g. error-type classifier or difficulty tagger on approved question bank + labeled mistakes (dozens–hundreds of examples), using a small open model or classic ML (sklearn + embeddings). Narrow task ≠ full generative AI.
- 3. Self-host an open model (ops, not “we invented it”).** Run llama3.1:8b on a lab GPU/VM. Still someone else’s weights; you own **deployment**. Honest line: “self-hosted open-weight LLM,” not “we trained it.”
- 4. Retrieval-augmented generation (RAG).** Generate feedback only from *your* approved explanations/topics — still uses an LLM API/local model, but the **knowledge is yours**. Stronger “we control quality” story.

Honest answer for the panel

“Building a frontier generative model wasn’t feasible or necessary. What we *can* own is task-specific models or a small fine-tune later. For generation and tutoring we use an open/commercial LLM as the inference engine, with our prompts, normalization, and human approval. The thesis contribution is the SRL/calibration platform, not founding a new LLM.”

Practical verdict for ExamiQ now

Goal	Feasible now?
Ship and defend the system	Use API/open LLM (what you do)
Say “we made an AI model”	Only if you add a small custom piece (classifier / fine-tune / RAG) and describe it narrowly
Replace Ollama entirely with a student-trained GPT	Not realistic in remaining thesis time

Bottom line: You don’t need to train GPT to have a valid AI thesis component. You *could* add a small custom model later for one task (e.g. error classification) if a panelist wants something “yours” — but redoing all generation as a homemade neural net is not a sensible path for this project.

